FURTHER MECHANICS / CENTRE OF MASS

Centre of Mass of Laminae by Integration

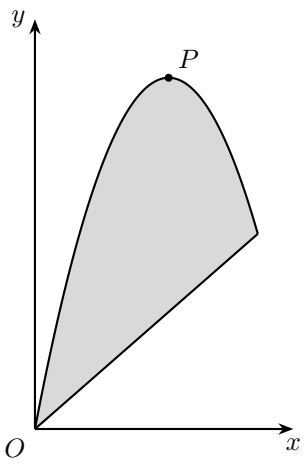
Worked solutions 10 questions · 115 marks

Instructions

Unless a question states otherwise, take $g = 9.8\,\mathrm{m\,s^{-2}}$ wherever a numerical value of g is required. Solutions also give final answers for $g = 10\,\mathrm{m\,s^{-2}}$ where these differ.

Questions

Question 1	*11 marks*	2
Question 2	*10 marks*	4
Question 3	*11 marks*	6
Question 4	*11 marks*	9
Question 5	*10 marks*	12
Question 6	*9 marks*	14
Question 7	*12 marks*	16
Question 8	*13 marks*	19
Question 9	*10 marks*	22
Question 10	*18 marks*	25



1. A uniform lamina occupies the region enclosed by the curve $y = 6x - x^2$ and the straight line $y = x$.
The unit of length on each axis is one centimetre.
The diagram above shows the lamina and the point $P(3, 9)$ at the maximum of the curve.

(a) Find the area of the lamina and use integration to find the coordinates of its centre of mass. Give your answers in exact form. **[8]**

(b) The lamina is freely suspended from P and hangs in equilibrium.
Find the acute angle between the vertical and the straight edge of the lamina, giving your answer to the nearest degree. **[3]**

Solution

(a) The boundaries intersect where

$$6x - x^2 = x \quad \Longrightarrow \quad x(5 - x) = 0 \quad \Longrightarrow \quad x = 0 \text{ or } x = 5$$

Regard the lamina as the region under the parabola for $0 \leq x \leq 5$, with the triangle beneath $y = x$ removed.

For a uniform lamina under a curve, the centre of mass formulae are

$$A = \int_a^b y \, \mathrm{d}x, \qquad \bar{x} = \frac{1}{A} \int_a^b xy \, \mathrm{d}x, \qquad \bar{y} = \frac{1}{2A} \int_a^b y^2 \, \mathrm{d}x$$

Using subscript 1 for the region under the parabola:

$$A_1 = \int_0^5 (6x - x^2) \, \mathrm{d}x = \left[3x^2 - \frac{x^3}{3} \right]_0^5 = \frac{100}{3} \, \mathrm{cm}^2$$

$$A_1 \bar{x}_1 = \int_0^5 (6x^2 - x^3) \, \mathrm{d}x = \left[2x^3 - \frac{x^4}{4} \right]_0^5 = \frac{375}{4} \, \mathrm{cm}^3$$

$$A_1 \bar{y}_1 = \frac{1}{2} \int_0^5 (36x^2 - 12x^3 + x^4) \, \mathrm{d}x$$

$$= \frac{1}{2} \left[12x^3 - 3x^4 + \frac{x^5}{5} \right]_0^5 = 125 \, \mathrm{cm}^3$$

The removed triangle has vertices $(0,0)$, $(5,0)$ and $(5,5)$. Its centroid coordinates are the means of the vertex coordinates, so

$$A_2 = \frac{1}{2}(5)(5) = \frac{25}{2}\,\text{cm}^2, \qquad \bar{x}_2 = \frac{0+5+5}{3} = \frac{10}{3}\,\text{cm}, \qquad \bar{y}_2 = \frac{0+0+5}{3} = \frac{5}{3}\,\text{cm}$$

Hence the area of the lamina is

$$A = A_1 - A_2 = \frac{100}{3} - \frac{25}{2} = \frac{125}{6}\,\text{cm}^2$$

For a uniform lamina made of pieces, the centre of mass formulae are

$$\bar{x} = \frac{\sum A_i \bar{x}_i}{\sum A_i}, \qquad \bar{y} = \frac{\sum A_i \bar{y}_i}{\sum A_i}$$

with the area of a removed piece counted as negative. Here this gives

$$\bar{x} = \frac{A_1\bar{x}_1 - A_2\bar{x}_2}{A_1 - A_2}, \qquad \bar{y} = \frac{A_1\bar{y}_1 - A_2\bar{y}_2}{A_1 - A_2}$$

Therefore

$$\bar{x} = \frac{\frac{375}{4} - \frac{25}{2}\times\frac{10}{3}}{\frac{125}{6}} = \frac{\frac{625}{12}}{\frac{125}{6}} = \frac{5}{2}\,\text{cm}$$

$$\bar{y} = \frac{125 - \frac{25}{2}\times\frac{5}{3}}{\frac{125}{6}} = \frac{\frac{625}{6}}{\frac{125}{6}} = 5\,\text{cm}$$

The centre of mass is $G\left(\frac{5}{2}, 5\right)$, with coordinates in centimetres.

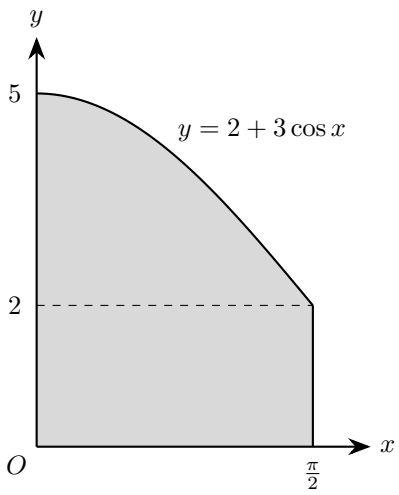
(b) In equilibrium, the centre of mass G lies vertically below the suspension point P. Thus the required angle is the acute angle between PG and the straight edge.

In the orientation shown in the question, let ϕ be the acute angle between PG and a vertical line. Using the horizontal and vertical separations of P and G:

$$\tan\phi = \frac{3 - \frac{5}{2}}{9 - 5} = \frac{1}{8}$$

The straight edge $y = x$ makes an angle of 45° with the vertical. Both it and PG lean to the same side of the downward vertical, so the required acute angle is

$$\theta = 45^\circ - \arctan\left(\frac{1}{8}\right) \approx 38^\circ \qquad \text{(nearest degree)}$$



2. A uniform lamina L is cut to form a decorative panel.
It occupies the region bounded by the curve $y = 2 + 3\cos x$, the x-axis and the vertical lines $x = 0$ and $x = \frac{\pi}{2}$, as shown in the diagram above.
Its centre of mass is $(\bar{x}, \bar{y})$.

(a) Using integration, show that

$$\bar{x} = \frac{\pi^2 + 6\pi - 12}{4(\pi + 3)}$$ **[5]**

(b) Find the exact value of $\bar{y}$. Hence show, without using decimal approximations, that the centre of mass lies above the line $y = 2$. **[5]**

Solution

(a) For a uniform lamina under a curve, the centre of mass formulae are

$$A = \int_a^b y \, \mathrm{d}x, \qquad \bar{x} = \frac{1}{A}\int_a^b xy \, \mathrm{d}x$$

Here the area is

$$A = \int_0^{\pi/2} (2 + 3\cos x) \, \mathrm{d}x = [2x + 3\sin x]_0^{\pi/2} = \pi + 3$$

For the term involving $x\cos x$, use integration by parts with $u = x$ and $\frac{\mathrm{d}v}{\mathrm{d}x} = \cos x$, so $\frac{\mathrm{d}u}{\mathrm{d}x} = 1$ and $v = \sin x$. This gives

$$\int_0^{\pi/2} x(2 + 3\cos x) \, \mathrm{d}x = \left[x^2 + 3x\sin x + 3\cos x\right]_0^{\pi/2}$$

$$= \frac{\pi^2}{4} + \frac{3\pi}{2} - 3$$

Dividing by the area:

$$\bar{x} = \frac{\frac{\pi^2}{4} + \frac{3\pi}{2} - 3}{\pi + 3} = \frac{\pi^2 + 6\pi - 12}{4(\pi + 3)}$$

(b) For a uniform lamina under a curve, the centre of mass formula is

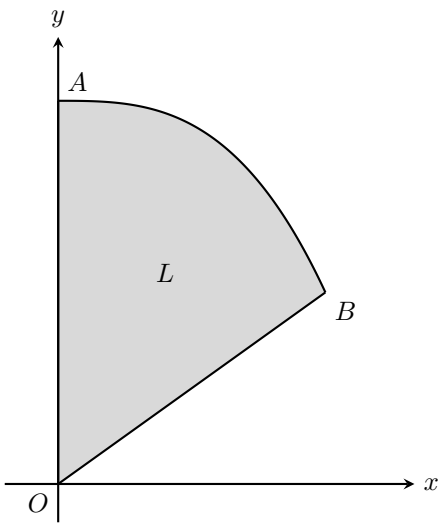
$$\bar{y} = \frac{1}{2A}\int_a^b y^2 \,\mathrm{d}x$$

Using $A = \pi + 3$ from part (a) and $\cos^2 x = \frac{1+\cos 2x}{2}$:

$$\begin{aligned}
\bar{y} &= \frac{1}{2(\pi+3)}\int_0^{\pi/2} (2+3\cos x)^2 \,\mathrm{d}x \\
&= \frac{1}{2(\pi+3)}\int_0^{\pi/2} \left(4 + 12\cos x + 9\cos^2 x\right) \mathrm{d}x \\
&= \frac{1}{2(\pi+3)}\int_0^{\pi/2} \left(\frac{17}{2} + 12\cos x + \frac{9}{2}\cos 2x\right) \mathrm{d}x \\
&= \frac{1}{2(\pi+3)}\left[\frac{17x}{2} + 12\sin x + \frac{9}{4}\sin 2x\right]_0^{\pi/2} \\
&= \frac{\frac{17\pi}{4} + 12}{2(\pi+3)} = \frac{17\pi + 48}{8(\pi+3)}
\end{aligned}$$

The centre of mass lies above the line $y = 2$, since

$$\bar{y} - 2 = \frac{17\pi + 48 - 16(\pi+3)}{8(\pi+3)} = \frac{\pi}{8(\pi+3)} > 0$$



3. The diagram above shows a region bounded by the y-axis, the straight line $y = x$ and the curve

$$y = 6 - \frac{x^3}{9}$$

The curve meets the y-axis at $A = (0, 6)$ and meets the line $y = x$ at $B = (3, 3)$.
The origin is O.

A uniform lamina L occupies the shaded region OAB.
The unit of length on each axis is one centimetre, and its centre of mass has coordinates $(\bar{x}, \bar{y})$.

(a) Use integration to find the area of L and show that

$$\bar{y} = \frac{118}{35}$$ **[5]**

(b) The lamina is freely suspended from B and hangs in equilibrium. Find the acute angle between the straight edge OB and the downward vertical, giving your answer in degrees to one decimal place. **[6]**

Solution

(a) Let R be the region under the curve and above the x-axis for $0 \leq x \leq 3$, and let T be the triangle with vertices O, $(3, 0)$ and B. The lamina L is obtained by removing T from R.

For a uniform lamina under a curve, the centre of mass formulae are

$$\mathcal{A} = \int_a^b y \, \mathrm{d}x, \qquad \bar{y} = \frac{1}{2\mathcal{A}} \int_a^b y^2 \, \mathrm{d}x$$

The areas are

$$\mathcal{A}_R = \int_0^3 \left(6 - \frac{x^3}{9}\right) \mathrm{d}x = \left[6x - \frac{x^4}{36}\right]_0^3 = \frac{63}{4} \, \mathrm{cm}^2$$

$$\mathcal{A}_T = \frac{1}{2} \times 3 \times 3 = \frac{9}{2} \, \mathrm{cm}^2$$

$$\mathcal{A}_L = \mathcal{A}_R - \mathcal{A}_T = \frac{63}{4} - \frac{9}{2} = \frac{45}{4} \, \mathrm{cm}^2$$

Writing the centre of mass of R as $(\bar{x}_R, \bar{y}_R)$, we obtain

$$\begin{aligned}\bar{y}_R &= \frac{1}{2(63/4)}\int_0^3 \left(6 - \frac{x^3}{9}\right)^2 \mathrm{d}x \\ &= \frac{2}{63}\int_0^3 \left(36 - \frac{4x^3}{3} + \frac{x^6}{81}\right)\mathrm{d}x \\ &= \frac{2}{63}\left[36x - \frac{x^4}{3} + \frac{x^7}{567}\right]_0^3 = \frac{132}{49}\,\mathrm{cm}\end{aligned}$$

The centroid of T has coordinates equal to the means of its vertex coordinates, so

$$\bar{y}_T = \frac{0+0+3}{3} = 1\,\mathrm{cm}$$

For a uniform lamina made of pieces, the centre of mass formula is

$$\bar{y} = \frac{\sum \mathcal{A}_i \bar{y}_i}{\sum \mathcal{A}_i}$$

with the area of a removed piece counted as negative. Hence

$$\begin{aligned}\bar{y} &= \frac{\mathcal{A}_R\bar{y}_R - \mathcal{A}_T\bar{y}_T}{\mathcal{A}_R - \mathcal{A}_T} \\ &= \frac{\frac{63}{4}\times\frac{132}{49} - \frac{9}{2}\times 1}{\frac{45}{4}} = \frac{118}{35}\,\mathrm{cm}\end{aligned}$$

(b) We first find $\bar{x}$. For a uniform lamina under a curve, the centre of mass formula is

$$\bar{x} = \frac{1}{\mathcal{A}}\int_a^b xy\,\mathrm{d}x$$

Applying this to R:

$$\begin{aligned}\bar{x}_R &= \frac{1}{63/4}\int_0^3 x\left(6 - \frac{x^3}{9}\right)\mathrm{d}x \\ &= \frac{4}{63}\left[3x^2 - \frac{x^5}{45}\right]_0^3 = \frac{48}{35}\,\mathrm{cm}\end{aligned}$$

For the triangle T, averaging the vertex coordinates gives

$$\bar{x}_T = \frac{0+3+3}{3} = 2\,\mathrm{cm}$$

For a uniform lamina made of pieces, the centre of mass formula is

$$\bar{x} = \frac{\sum \mathcal{A}_i \bar{x}_i}{\sum \mathcal{A}_i}$$

with the area of a removed piece counted as negative. Hence

$$\begin{aligned}\bar{x} &= \frac{\mathcal{A}_R\bar{x}_R - \mathcal{A}_T\bar{x}_T}{\mathcal{A}_R - \mathcal{A}_T} \\ &= \frac{\frac{63}{4}\times\frac{48}{35} - \frac{9}{2}\times 2}{\frac{45}{4}} = \frac{28}{25}\,\mathrm{cm}\end{aligned}$$

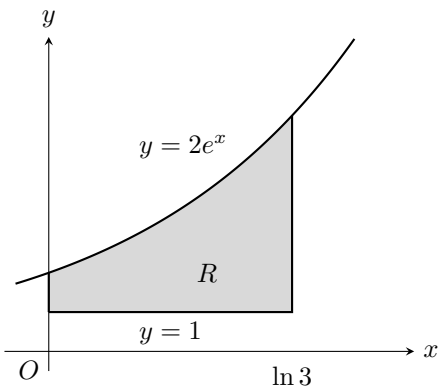
Thus the centre of mass is $G\left(\frac{28}{25}, \frac{118}{35}\right)$. When the lamina hangs in equilibrium, G lies vertically below the suspension point B. The required angle is therefore $\angle OBG$.

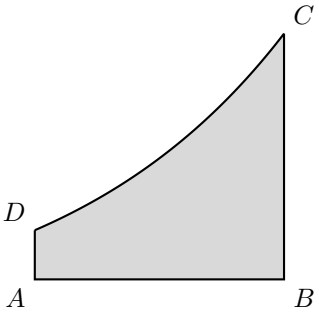
In the original diagram, G is to the left of and above $B(3,3)$. Let ϕ be the acute angle between BG and the leftward horizontal through B. Then

$$\begin{aligned}\tan\phi &= \frac{\bar{y}-3}{3-\bar{x}} \\ &= \frac{\frac{118}{35}-3}{3-\frac{28}{25}} = \frac{65}{329} \\ \phi &= \arctan\left(\frac{65}{329}\right)\end{aligned}$$

Since OB lies on $y=x$, the direction from B towards O is 45° below the leftward horizontal. Hence the required acute angle is

$$\angle OBG = 45^\circ + \phi \approx 56.2^\circ \qquad (1 \text{ d.p.})$$





4. The shaded region R is bounded by the curve $y = 2e^x$, the horizontal line $y = 1$, the y-axis and the line $x = \ln 3$.
The unit of length on each axis is one metre.

A uniform lamina occupies R, and its centre of mass has coordinates $(\bar{x}, \bar{y})$.

(a) Use integration to find the area of R and show that

$$\bar{y} = \frac{16 - \ln 3}{8 - 2\ln 3} \qquad \textbf{[5]}$$

The second diagram above shows the same lamina, with vertices $A = (0, 1)$, $B = (\ln 3, 1)$, $C = (\ln 3, 6)$ and $D = (0, 2)$.

The lamina is freely suspended from D and hangs in equilibrium.
Let θ be the angle between the directed segment from D to A and the downward vertical, where $0° \leq \theta \leq 180°$.

(b) Use integration to find $\bar{x}$ in exact form. Hence find θ, giving your answer to one decimal place. **[6]**

Solution

(a) Regard R as the region S beneath $y = 2e^x$ and above the x-axis, for $0 \leq x \leq \ln 3$, with the rectangle T of height 1 removed.

For a uniform lamina under a curve, the centre of mass formulae are

$$\mathcal{A} = \int_a^b y \, \mathrm{d}x, \qquad \bar{y} = \frac{1}{2\mathcal{A}} \int_a^b y^2 \, \mathrm{d}x$$

The areas of S, T and R are therefore

$$\mathcal{A}_S = \int_0^{\ln 3} 2e^x \,\mathrm{d}x = [2e^x]_0^{\ln 3} = 4\,\mathrm{m}^2$$

$$\mathcal{A}_T = \ln 3\,\mathrm{m}^2$$

$$\mathcal{A} = \mathcal{A}_S - \mathcal{A}_T = (4 - \ln 3)\,\mathrm{m}^2$$

The vertical coordinate of the centre of mass of S is

$$\bar{y}_S = \frac{1}{2(4)} \int_0^{\ln 3} 4e^{2x} \,\mathrm{d}x$$

$$= \frac{1}{8}\left[2e^{2x}\right]_0^{\ln 3} = 2\,\mathrm{m}$$

The rectangle has $\bar{y}_T = \frac{1}{2}\,\mathrm{m}$. For a uniform lamina made of pieces, the centre of mass formula is

$$\bar{y} = \frac{\sum \mathcal{A}_i \bar{y}_i}{\sum \mathcal{A}_i}$$

with the area of a removed piece counted as negative. Hence

$$\bar{y} = \frac{\mathcal{A}_S \bar{y}_S - \mathcal{A}_T \bar{y}_T}{\mathcal{A}_S - \mathcal{A}_T}$$

$$= \frac{4(2) - (\ln 3)\left(\frac{1}{2}\right)}{4 - \ln 3}$$

$$= \frac{16 - \ln 3}{8 - 2\ln 3}\,\mathrm{m}$$

(b) For a uniform lamina under a curve, the centre of mass formula is

$$\bar{x} = \frac{1}{\mathcal{A}} \int_a^b xy \,\mathrm{d}x$$

For S, integrate by parts, taking $u = x$ and $\frac{\mathrm{d}v}{\mathrm{d}x} = 2e^x$, so that $\frac{\mathrm{d}u}{\mathrm{d}x} = 1$ and $v = 2e^x$:

$$\mathcal{A}_S \bar{x}_S = \int_0^{\ln 3} 2xe^x \,\mathrm{d}x$$

$$= [2xe^x]_0^{\ln 3} - \int_0^{\ln 3} 2e^x \,\mathrm{d}x$$

$$= [2(x-1)e^x]_0^{\ln 3}$$

$$= (6\ln 3 - 4)\,\mathrm{m}^3$$

The rectangle has $\bar{x}_T = \frac{\ln 3}{2}\,\mathrm{m}$. For a uniform lamina made of pieces, the centre of mass formula is

$$\bar{x} = \frac{\sum \mathcal{A}_i \bar{x}_i}{\sum \mathcal{A}_i}$$

with the area of a removed piece counted as negative. Hence

$$\begin{aligned}\bar{x} &= \frac{\mathcal{A}_S \bar{x}_S - \mathcal{A}_T \bar{x}_T}{\mathcal{A}_S - \mathcal{A}_T} \\ &= \frac{6\ln 3 - 4 - \frac{1}{2}(\ln 3)^2}{4 - \ln 3} \\ &= \frac{12\ln 3 - 8 - (\ln 3)^2}{8 - 2\ln 3}\ \text{m}\end{aligned}$$

Let G be the centre of mass. In equilibrium, G lies vertically below the suspension point D, so θ is the angle between the directed segments DA and DG.

In the original orientation, $\bar{x} > 0$, and part (a) gives

$$\bar{y} - 2 = \frac{16 - \ln 3 - 2(8 - 2\ln 3)}{8 - 2\ln 3} = \frac{3\ln 3}{8 - 2\ln 3} > 0$$

Thus G is above and to the right of D, whereas DA points downwards. Hence θ is obtuse, and the right-angled triangle formed by the horizontal and vertical displacements from D to G gives

$$\begin{aligned}\tan(180^\circ - \theta) &= \frac{\bar{x}}{\bar{y} - 2} \\ &= \frac{12\ln 3 - 8 - (\ln 3)^2}{3\ln 3} \\ \theta &= 180^\circ - \arctan\left(\frac{12\ln 3 - 8 - (\ln 3)^2}{3\ln 3}\right) \\ &\approx 129.7^\circ \qquad (1\text{ d.p.})\end{aligned}$$

5. A uniform lamina occupies the region bounded by the curve

$$y = \frac{k}{x+1}$$

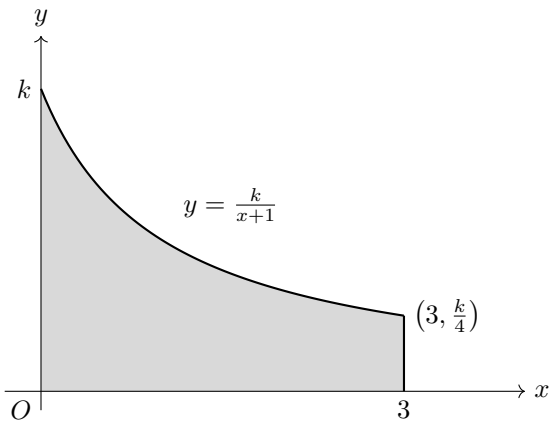
the x-axis and the vertical lines $x = 0$ and $x = 3$, where k is a positive constant.

(i) Find the exact x-coordinate of the centre of mass of the lamina. **[5]**

(ii) The centre of mass lies on the line $y = x$.
Find the exact value of k and hence the exact area of the lamina. **[5]**

Solution

(i) A schematic diagram of the lamina is:



For a uniform lamina under a curve, the centre of mass formulae are

$$A = \int_a^b y \, \mathrm{d}x, \qquad \bar{x} = \frac{1}{A}\int_a^b xy \, \mathrm{d}x$$

The area of this lamina is

$$\begin{aligned} A &= \int_0^3 \frac{k}{x+1} \, \mathrm{d}x \\ &= k[\ln(x+1)]_0^3 \\ &= k \ln 4 \end{aligned}$$

Using $\frac{x}{x+1} = 1 - \frac{1}{x+1}$:

$$\begin{aligned} \int_0^3 xy \, \mathrm{d}x &= k \int_0^3 \frac{x}{x+1} \, \mathrm{d}x \\ &= k \int_0^3 \left(1 - \frac{1}{x+1}\right) \mathrm{d}x \\ &= k[x - \ln(x+1)]_0^3 \\ &= k(3 - \ln 4) \end{aligned}$$

Therefore

$$\bar{x} = \frac{k(3 - \ln 4)}{k \ln 4} = \frac{3 - \ln 4}{\ln 4}$$

(ii) For a uniform lamina under a curve, the centre of mass formula is

$$\bar{y} = \frac{1}{2A} \int_a^b y^2 \, \mathrm{d}x$$

Using $A = k \ln 4$ from part (i):

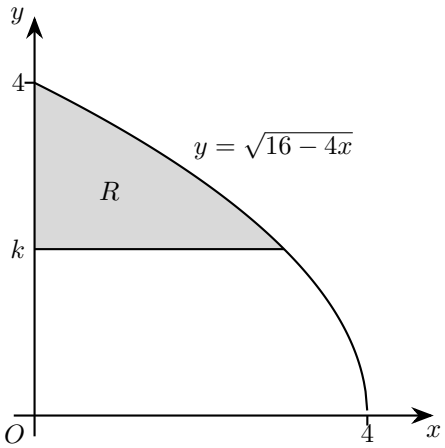
$$\begin{aligned}
\bar{y} &= \frac{1}{2k \ln 4} \int_0^3 \frac{k^2}{(x+1)^2} \, \mathrm{d}x \\
&= \frac{k}{2 \ln 4} \left[-\frac{1}{x+1} \right]_0^3 \\
&= \frac{k}{2 \ln 4} \left(1 - \frac{1}{4} \right) \\
&= \frac{3k}{8 \ln 4}
\end{aligned}$$

Since the centre of mass lies on $y = x$, we have $\bar{y} = \bar{x}$. Thus

$$\begin{aligned}
\frac{3k}{8 \ln 4} &= \frac{3 - \ln 4}{\ln 4} \\
3k &= 8(3 - \ln 4) \\
k &= \frac{8(3 - \ln 4)}{3}
\end{aligned}$$

Hence the exact area is

$$A = k \ln 4 = \frac{8(3 - \ln 4) \ln 4}{3}$$



6. The shaded region R is bounded by the curve $y = \sqrt{16 - 4x}$, the y-axis and the horizontal line $y = k$, where $0 < k < 4$.
A uniform lamina occupies R.

(a) Using integration, determine the y-coordinate of the centre of mass of the lamina in terms of k. Simplify your expression fully. **[6]**

(b) The centre of mass lies on the line $y = 3$. Find the exact value of k, explaining why any other root of your equation must be rejected. **[3]**

Solution

(a) Rearranging the curve equation to express x in terms of y gives

$$x = 4 - \frac{y^2}{4}$$

For a uniform lamina against the y-axis, the centre of mass formulae are

$$A = \int_a^b x \, \mathrm{d}y, \qquad \bar{y} = \frac{1}{A} \int_a^b xy \, \mathrm{d}y$$

The area is therefore

$$\begin{aligned} A &= \int_k^4 \left(4 - \frac{y^2}{4}\right) \mathrm{d}y \\ &= \left[4y - \frac{y^3}{12}\right]_k^4 \\ &= \frac{32}{3} - 4k + \frac{k^3}{12} \\ &= \frac{k^3 - 48k + 128}{12} \end{aligned}$$

Using the centre of mass formula:

$$\begin{aligned} A\bar{y} &= \int_k^4 y\left(4 - \frac{y^2}{4}\right) \mathrm{d}y \\ &= \left[2y^2 - \frac{y^4}{16}\right]_k^4 \\ &= 16 - 2k^2 + \frac{k^4}{16} \\ &= \frac{(k^2-16)^2}{16} \end{aligned}$$

Dividing by A, factorising and cancelling $(k-4)^2$, which is non-zero since $0 < k < 4$, gives

$$\begin{aligned} \bar{y} &= \frac{3(k^2-16)^2}{4(k^3-48k+128)} \\ &= \frac{3(k-4)^2(k+4)^2}{4(k-4)^2(k+8)} \\ &= \frac{3(k+4)^2}{4(k+8)} \end{aligned}$$

(b) Since the centre of mass lies on $y = 3$, part (a) gives

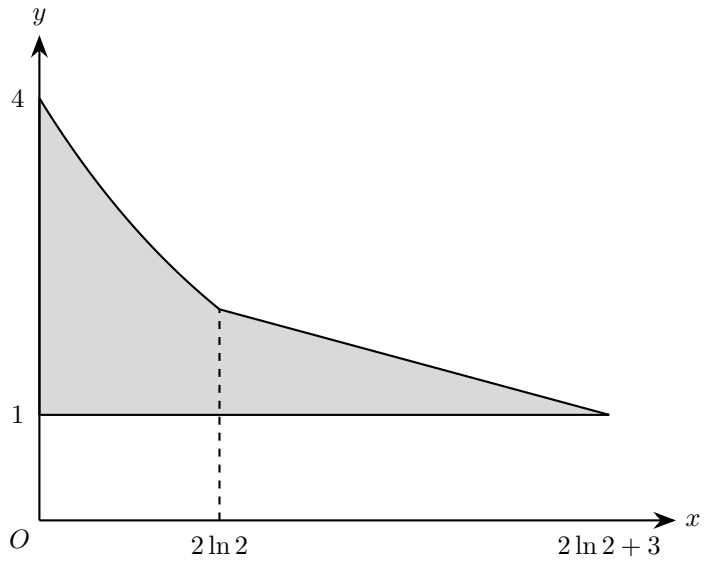
$$\begin{aligned} \frac{3(k+4)^2}{4(k+8)} &= 3 \\ (k+4)^2 &= 4(k+8) \\ k^2 + 4k - 16 &= 0 \\ (k+2)^2 &= 20 \\ k &= -2 \pm 2\sqrt{5} \end{aligned}$$

The root $-2 - 2\sqrt{5}$ is negative, so it is rejected because $0 < k < 4$. The other root lies between 0 and 4, so

$$k = 2(\sqrt{5} - 1)$$

7. A uniform lamina occupies the region between the curve $y = 4e^{-x/2}$ and the line $y = 1$, for $0 \leq x \leq 2\ln 2$.

(i) Using integration, find the exact area of this lamina and the exact y-coordinate of its centre of mass. **[5]**



A triangular piece of lamina with the same uniform mass per unit area is attached to its right-hand edge. The resulting lamina is shown in the diagram above.
Its lower boundary is $y = 1$ and its upper boundary is $y = f(x)$, where

$$f(x) = \begin{cases} 4e^{-x/2}, & 0 \leq x \leq 2\ln 2 \\ 2 - \dfrac{x - 2\ln 2}{3}, & 2\ln 2 \leq x \leq 2\ln 2 + 3 \end{cases}$$

Its left-hand boundary lies on the y-axis.

(ii) Using your area from part (i), find the exact x-coordinate of the centre of mass of the resulting lamina. **[7]**

Solution

(i) Regard the lamina as the region under $y = 4e^{-x/2}$ down to the x-axis, with a rectangle removed. Denote these two regions by C and R, respectively.

For a uniform lamina under a curve, the centre of mass formulae are

$$A = \int_a^b y\,\mathrm{d}x, \qquad \bar{y} = \frac{1}{2A}\int_a^b y^2\,\mathrm{d}x$$

The area of C is

$$\begin{aligned} A_C &= \int_0^{2\ln 2} 4e^{-x/2}\,\mathrm{d}x \\ &= \left[-8e^{-x/2}\right]_0^{2\ln 2} \\ &= -4 + 8 = 4 \end{aligned}$$

The removed rectangle has width $2\ln 2$ and height 1, so the required area is

$$A = A_C - A_R = 4 - 2\ln 2$$

Applying the centre of mass formula to C:

$$\begin{aligned}\bar{y}_C &= \frac{1}{2(4)}\int_0^{2\ln 2} 16e^{-x}\,\mathrm{d}x \\ &= \frac{1}{8}\left[-16e^{-x}\right]_0^{2\ln 2} \\ &= \frac{1}{8}(-4+16) = \frac{3}{2}\end{aligned}$$

The rectangle has $\bar{y}_R = \frac{1}{2}$. For a uniform lamina made of pieces, the centre of mass formula is

$$\bar{y} = \frac{\sum A_i \bar{y}_i}{\sum A_i}$$

with the area of a removed piece counted as negative. Hence

$$\begin{aligned}\bar{y} &= \frac{A_C\bar{y}_C - A_R\bar{y}_R}{A_C - A_R} \\ &= \frac{4\left(\frac{3}{2}\right) - (2\ln 2)\left(\frac{1}{2}\right)}{4 - 2\ln 2} \\ &= \frac{6 - \ln 2}{4 - 2\ln 2}\end{aligned}$$

(ii) First find the horizontal position of the centre of mass of the original lamina, using the same regions C and R.

For a uniform lamina under a curve, the centre of mass formula is

$$\bar{x} = \frac{1}{A}\int_a^b xy\,\mathrm{d}x$$

To integrate $4xe^{-x/2}$ by parts, take $u = x$ and $\frac{\mathrm{d}v}{\mathrm{d}x} = 4e^{-x/2}$, so $\frac{\mathrm{d}u}{\mathrm{d}x} = 1$ and $v = -8e^{-x/2}$. Thus

$$\begin{aligned}A_C\bar{x}_C &= \int_0^{2\ln 2} 4xe^{-x/2}\,\mathrm{d}x \\ &= \left[-8xe^{-x/2}\right]_0^{2\ln 2} + 8\int_0^{2\ln 2} e^{-x/2}\,\mathrm{d}x \\ &= \left[-8(x+2)e^{-x/2}\right]_0^{2\ln 2} \\ &= -4(2\ln 2 + 2) + 16 \\ &= 8 - 8\ln 2\end{aligned}$$

The rectangle has $\bar{x}_R = \ln 2$. For a uniform lamina made of pieces, the centre of mass formula is

$$\bar{x} = \frac{\sum A_i \bar{x}_i}{\sum A_i}$$

with the area of a removed piece counted as negative. Using the area $A = 4 - 2\ln 2$ from part (i), the original lamina therefore has

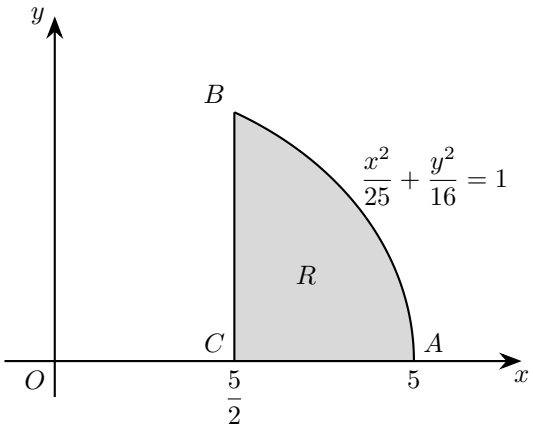
$$\begin{aligned}\bar{x}_0 &= \frac{A_C\bar{x}_C - A_R\bar{x}_R}{A} \\ &= \frac{8 - 8\ln 2 - (2\ln 2)(\ln 2)}{4 - 2\ln 2} \\ &= \frac{8 - 8\ln 2 - 2(\ln 2)^2}{4 - 2\ln 2}\end{aligned}$$

The attached triangle has base 3 and height 1. Its centroid's x-coordinate is the mean of its vertices' x-coordinates, so

$$A_T = \frac{1}{2}(3)(1) = \frac{3}{2}, \qquad \bar{x}_T = \frac{2\ln 2 + 2\ln 2 + (2\ln 2 + 3)}{3} = 2\ln 2 + 1$$

Since both pieces have the same uniform mass per unit area, the same formula applies to the original lamina and the attached triangle:

$$\begin{aligned}\bar{x} &= \frac{A\bar{x}_0 + A_T\bar{x}_T}{A + A_T} \\ &= \frac{8 - 8\ln 2 - 2(\ln 2)^2 + \frac{3}{2}(2\ln 2 + 1)}{4 - 2\ln 2 + \frac{3}{2}} \\ &= \frac{\frac{19}{2} - 5\ln 2 - 2(\ln 2)^2}{\frac{11}{2} - 2\ln 2} \\ &= \frac{19 - 10\ln 2 - 4(\ln 2)^2}{11 - 4\ln 2}\end{aligned}$$



8. A uniform lamina occupies the shaded region R, bounded by the upper arc of the ellipse

$$\frac{x^2}{25}+\frac{y^2}{16}=1$$

the line $x=\frac{5}{2}$ and the x-axis.

The points C and A have coordinates $\left(\frac{5}{2},0\right)$ and $(5,0)$ respectively.
The point B is the upper intersection of the ellipse with $x=\frac{5}{2}$.
The unit of length on each axis is one metre.

(a) Using the substitution $x=5\sin t$, show by integration that the area of R is

$$\frac{5}{6}\left(4\pi-3\sqrt{3}\right)\text{ m}^2 \qquad \textbf{[4]}$$

(b) Use integration to find the exact coordinates $(\bar{x},\bar{y})$ of the centre of mass of the lamina. **[6]**

(c) The lamina is freely suspended from A and hangs in equilibrium. Find the acute angle between AC and the downward vertical, giving your answer in degrees to one decimal place. **[3]**

Solution

(a) The upper arc of the ellipse has equation

$$y=4\sqrt{1-\frac{x^2}{25}}$$

Using the given substitution $x=5\sin t$, we have $\mathrm{d}x=5\cos t\,\mathrm{d}t$ and $y=4\cos t$. The limits $x=\frac{5}{2}$ and $x=5$ correspond to $t=\frac{\pi}{6}$ and $t=\frac{\pi}{2}$, respectively.

The area $\mathcal{A}$ is therefore

$$\begin{aligned}\mathcal{A} &= \int_{5/2}^{5} y\,\mathrm{d}x \\ &= 20\int_{\pi/6}^{\pi/2} \cos^2 t\,\mathrm{d}t \\ &= 10\int_{\pi/6}^{\pi/2} (1+\cos 2t)\,\mathrm{d}t \\ &= 10\left[t + \frac{1}{2}\sin 2t\right]_{\pi/6}^{\pi/2} \\ &= \frac{10\pi}{3} - \frac{5\sqrt{3}}{2} \\ &= \frac{5}{6}\left(4\pi - 3\sqrt{3}\right)\mathrm{m}^2\end{aligned}$$

(b) A thin vertical strip has area $y\,\mathrm{d}x$ and its centre is at $\left(x, \frac{y}{2}\right)$. For a uniform lamina under a curve, the centre of mass formulae are

$$\bar{x} = \frac{1}{\mathcal{A}}\int_a^b xy\,\mathrm{d}x, \qquad \bar{y} = \frac{1}{2\mathcal{A}}\int_a^b y^2\,\mathrm{d}x$$

Writing $y = \frac{4}{5}\sqrt{25-x^2}$ and integrating by the reverse chain rule:

$$\begin{aligned}\int_{5/2}^{5} xy\,\mathrm{d}x &= \frac{4}{5}\int_{5/2}^{5} x(25-x^2)^{1/2}\,\mathrm{d}x \\ &= \left[-\frac{4}{15}(25-x^2)^{3/2}\right]_{5/2}^{5} \\ &= \frac{25\sqrt{3}}{2}\,\mathrm{m}^3\end{aligned}$$

For the other coordinate, $y^2 = 16\left(1 - \frac{x^2}{25}\right)$, so

$$\begin{aligned}\int_{5/2}^{5} \frac{y^2}{2}\,\mathrm{d}x &= 8\int_{5/2}^{5}\left(1 - \frac{x^2}{25}\right)\mathrm{d}x \\ &= 8\left[x - \frac{x^3}{75}\right]_{5/2}^{5} \\ &= \frac{25}{3}\,\mathrm{m}^3\end{aligned}$$

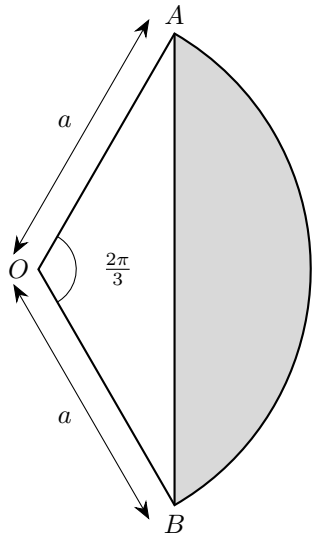
Using the area from part (a), the exact coordinates are

$$(\bar{x}, \bar{y}) = \left(\frac{25\sqrt{3}}{2\mathcal{A}}, \frac{25}{3\mathcal{A}}\right) = \left(\frac{15\sqrt{3}}{4\pi - 3\sqrt{3}}, \frac{10}{4\pi - 3\sqrt{3}}\right)\mathrm{m}$$

(c) Let G be the centre of mass and let θ be the required acute angle. In equilibrium, G lies vertically below A, so θ is the angle between AC and AG.

In the original coordinates, the horizontal and vertical separations of A and G are $5-\bar{x}$ and $\bar{y}$. Hence

$$\begin{aligned}
\tan\theta &= \frac{\bar{y}}{5-\bar{x}} \\
&= \frac{10}{5(4\pi-3\sqrt{3})-15\sqrt{3}} \\
&= \frac{1}{2\pi-3\sqrt{3}} \\
\theta &= \arctan\left(\frac{1}{2\pi-3\sqrt{3}}\right) \approx 42.6^\circ \qquad (1 \text{ d.p.})
\end{aligned}$$



9. A thin uniform metal plate has the shape of the shaded circular segment in the diagram above.
It is bounded by the chord AB and the minor arc AB of a circle with centre O and radius a, where $a > 0$.
The angle AOB is $\frac{2\pi}{3}$.

The plate is modelled as a uniform lamina with constant mass per unit area σ, where $\sigma > 0$.

Take O as the origin and the positive x-axis along the bisector of angle AOB, directed towards the minor arc.
Thus the chord AB lies on $x = \frac{a}{2}$.

(a) Using integration, find the mass of the plate in terms of σ and a. Give your answer in exact form. **[5]**

(b) Using integration to calculate the first moment of the plate about the y-axis, find the exact distance of its centre of mass from O. Explain why the centre of mass lies on the x-axis. **[5]**

Solution

(a) The upper endpoint A has height

$$h = a\sin\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}a}{2}$$

Consider the region R_1 between the curve $x = \sqrt{a^2 - y^2}$, the y-axis and the lines $y = 0$ and $y = h$.
The upper half of the plate is obtained by removing the rectangle R_2, of width $\frac{a}{2}$ and height h, from R_1.

For a region between a curve and the y-axis, the area formula gives

$$\mathcal{A}_1 = \int_0^h x\,\mathrm{d}y = \int_0^h \sqrt{a^2 - y^2}\,\mathrm{d}y$$

Use $y = a\sin t$, so $\mathrm{d}y = a\cos t\,\mathrm{d}t$. The limits become $t = 0$ and $t = \frac{\pi}{3}$, and $\sqrt{a^2 - y^2} = a\cos t$. Hence

$$\begin{aligned}\mathcal{A}_1 &= a^2\int_0^{\pi/3}\cos^2 t\,\mathrm{d}t\\ &= \frac{a^2}{2}\int_0^{\pi/3}(1+\cos 2t)\,\mathrm{d}t\\ &= \frac{a^2}{2}\left[t + \frac{\sin 2t}{2}\right]_0^{\pi/3}\\ &= a^2\left(\frac{\pi}{6} + \frac{\sqrt{3}}{8}\right)\end{aligned}$$

The rectangle has area $\mathcal{A}_2 = \frac{ah}{2} = \frac{\sqrt{3}a^2}{4}$. Doubling the upper-half area and multiplying by the mass per unit area gives

$$\begin{aligned}M &= 2\sigma(\mathcal{A}_1 - \mathcal{A}_2)\\ &= 2\sigma a^2\left(\frac{\pi}{6} + \frac{\sqrt{3}}{8} - \frac{\sqrt{3}}{4}\right)\\ &= \frac{\sigma a^2(4\pi - 3\sqrt{3})}{12}\end{aligned}$$

(b) The plate is symmetric about the x-axis, so its centre of mass lies on that axis and $\bar{y} = 0$.

For a uniform lamina against the y-axis, the centre of mass formula is

$$\bar{x} = \frac{1}{2\mathcal{A}}\int_c^d x^2\,\mathrm{d}y$$

The rectangle R_2 has $\bar{x}_2 = \frac{a}{4}$. For a uniform lamina made of pieces, the centre of mass formula is

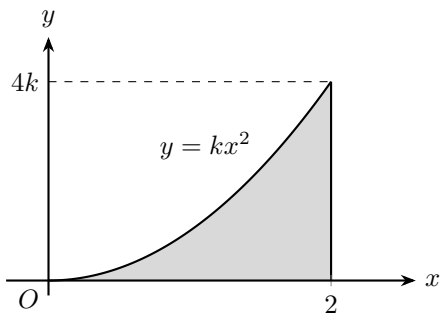
$$\bar{x} = \frac{\sum \mathcal{A}_i\bar{x}_i}{\sum \mathcal{A}_i}$$

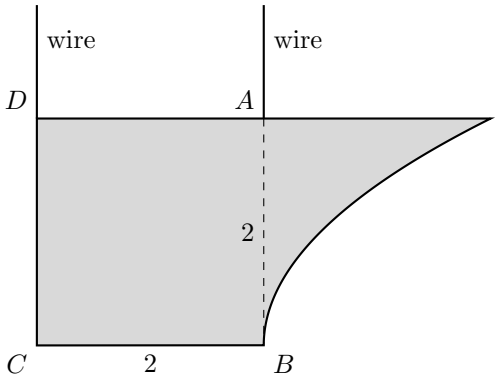
with the area of a removed piece counted as negative. Multiplied by σ, its numerator is the first moment about the y-axis. Applying the formula for a lamina against the y-axis to R_1, subtracting the rectangle's contribution and doubling to include the lower half gives

$$\begin{aligned}Q_y &= 2\sigma(\mathcal{A}_1\bar{x}_1 - \mathcal{A}_2\bar{x}_2)\\ &= 2\sigma\left(\frac{1}{2}\int_0^h (a^2 - y^2)\,\mathrm{d}y - \frac{ah}{2}\cdot\frac{a}{4}\right)\\ &= \sigma\left[a^2 y - \frac{y^3}{3}\right]_0^h - \frac{\sigma a^2 h}{4}\\ &= \sigma\left(\frac{3a^2h}{4} - \frac{h^3}{3}\right)\\ &= \sigma\left(\frac{3\sqrt{3}a^3}{8} - \frac{\sqrt{3}a^3}{8}\right)\\ &= \frac{\sqrt{3}\sigma a^3}{4}\end{aligned}$$

Since the plate lies to the right of O, the required distance is $\bar{x}$. Using the mass from part (a):

$$\bar{x} = \frac{Q_y}{M} = \frac{\dfrac{\sqrt{3}\sigma a^3}{4}}{\dfrac{\sigma a^2(4\pi - 3\sqrt{3})}{12}} = \frac{3\sqrt{3}a}{4\pi - 3\sqrt{3}}$$





10. A uniform lamina occupies the region bounded by the curve $y = kx^2$, the x-axis and the line $x = 2$, where $k > 0$, as shown in the first diagram above.

(i) Use integration to find both coordinates of the centre of mass of this lamina in terms of k. **[8]**

A sign is made from a uniform sheet by joining a square $ABCD$ of side length 2 to a curved piece having the shape of the lamina in part (i).
The sheet has the same thickness and density throughout.

For the sign, take C as the origin, with CB along the positive x-axis and CD along the positive y-axis.
Thus $A = (2, 2)$, $B = (2, 0)$ and $D = (0, 2)$.
The curved piece is obtained by reflecting the lamina in part (i) in the line $y = x$ and then translating it 2 units to the right.
Its curved edge therefore has equation

$$x = 2 + ky^2, \qquad 0 \leq y \leq 2$$

The complete sign is shown in the second diagram above.

(ii) Find both coordinates of the centre of mass of the sign in terms of k. **[5]**

The sign is held in equilibrium in a vertical plane by two vertical wires attached at A and D.
The tension in the wire at A is twice the tension in the wire at D.

(iii) Find the exact value of k. **[5]**

Solution

(i) A vertical strip of width $\mathrm{d}x$ under the curve has area $y\,\mathrm{d}x$ and centre at $\left(x, \frac{1}{2}y\right)$. For a uniform lamina under a curve, the centre of mass formulae are

$$\mathcal{A} = \int_a^b y\,\mathrm{d}x, \qquad \bar{x}_L = \frac{1}{\mathcal{A}}\int_a^b xy\,\mathrm{d}x, \qquad \bar{y}_L = \frac{1}{2\mathcal{A}}\int_a^b y^2\,\mathrm{d}x$$

Here $y = kx^2$, with $0 \le x \le 2$. The area is

$$\begin{aligned}\mathcal{A} &= \int_0^2 kx^2\,\mathrm{d}x \\ &= k\left[\frac{x^3}{3}\right]_0^2 = \frac{8k}{3}\end{aligned}$$

Applying the centre of mass formulae:

$$\begin{aligned}\bar{x}_L &= \frac{1}{8k/3}\int_0^2 kx^3\,\mathrm{d}x \\ &= \frac{k}{8k/3}\left[\frac{x^4}{4}\right]_0^2 = \frac{4k}{8k/3} = \frac{3}{2} \\ \bar{y}_L &= \frac{1}{8k/3}\cdot\frac{1}{2}\int_0^2 k^2x^4\,\mathrm{d}x \\ &= \frac{k^2}{2(8k/3)}\left[\frac{x^5}{5}\right]_0^2 = \frac{16k^2/5}{8k/3} = \frac{6k}{5}\end{aligned}$$

(ii) Reflection in $y = x$ interchanges the centre's coordinates, and translation 2 units to the right adds 2 to its x-coordinate. Using part (i), the curved piece has area and centre

$$\mathcal{A}_p = \frac{8k}{3}, \qquad (\bar{x}_p, \bar{y}_p) = \left(2 + \frac{6k}{5}, \frac{3}{2}\right)$$

The square has area $\mathcal{A}_s = 4$ and centre $(\bar{x}_s, \bar{y}_s) = (1, 1)$. Since the thickness and density are the same throughout, masses are proportional to areas. For a uniform lamina made of pieces, the centre of mass formulae are

$$\bar{x} = \frac{\sum \mathcal{A}_i\bar{x}_i}{\sum \mathcal{A}_i}, \qquad \bar{y} = \frac{\sum \mathcal{A}_i\bar{y}_i}{\sum \mathcal{A}_i}$$

Summing over the square and the curved piece:

$$\begin{aligned}\bar{x} &= \frac{4 + \frac{8k}{3}\left(2 + \frac{6k}{5}\right)}{4 + \frac{8k}{3}} \\ &= \frac{4 + \frac{16k}{3} + \frac{16k^2}{5}}{4 + \frac{8k}{3}} = \frac{12k^2 + 20k + 15}{10k + 15} \\ \bar{y} &= \frac{4 + \frac{8k}{3}\cdot\frac{3}{2}}{4 + \frac{8k}{3}} = \frac{4 + 4k}{4 + \frac{8k}{3}} = \frac{3(k+1)}{2k+3}\end{aligned}$$

(iii) Let the tensions be T_A and T_D, and let the sign's weight be W. Vertical equilibrium gives

$$W = T_A + T_D = 2T_D + T_D = 3T_D$$

Taking moments about A, the tension at D has horizontal distance 2, while the weight acts vertically through $x = \bar{x}$. Thus

$$\begin{aligned} W(2 - \bar{x}) &= 2T_D \\ 3T_D(2 - \bar{x}) &= 2T_D \\ 2 - \bar{x} &= \frac{2}{3} \\ \bar{x} &= \frac{4}{3} \end{aligned}$$

Using the x-coordinate from part (ii):

$$\begin{aligned} \frac{12k^2 + 20k + 15}{10k + 15} &= \frac{4}{3} \\ 3(12k^2 + 20k + 15) &= 4(10k + 15) \\ 36k^2 + 60k + 45 &= 40k + 60 \\ 36k^2 + 20k - 15 &= 0 \\ k &= \frac{-20 \pm \sqrt{20^2 - 4(36)(-15)}}{72} \\ &= \frac{-5 \pm 4\sqrt{10}}{18} \end{aligned}$$

Since $k > 0$, the required value is

$$k = \frac{4\sqrt{10} - 5}{18}$$